Newsletter

Europe's green tech renaissance: seizing opportunities while guarding against over-reliance on China

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Authors Alicia García-Herrero

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In the race to net-zero emissions, China's ambition to transform itself into an 'electrostate' – a nation deriving economic and political power from abundant, cheap electricity – presents both a model and a cautionary tale for the European Union. Unlike traditional petrostates like Saudi Arabia, which wield influence through oil exports, an electrostate leverages cheap electricity to fuel its own industrial might, producing everything from green steel to solar panels. As energy in its power grid lags behind many European countries and the share of emissions from its massive industrial sector remain stubbornly high at 50%. Yet, China's strides in electric vehicles (EVs) and clean tech exports signal a strategic pivot that could increase China's global competitive advantage even further.

The EU, often portrayed as lagging hopelessly behind, has more reason for optimism than sceptics admit. Based on Bruegel's European Clean Tech Tracker, the data reveals a continent quietly building a robust clean tech ecosystem. Take batteries and EVs: the EU boasts significant manufacturing capacity across Hungary, Poland, France and Germany. Facilities for cell fabrication – the most capital-intensive part of battery production – are scaling up, supplying European automakers and creating jobs. Germany alone exports billions of euros worth of EVs annually, demonstrating that the continent isn't just consuming green tech but producing and selling it globally. In wind energy and heat pumps, the EU maintains strong domestic capabilities, while the solar sector employs hundreds of thousands – not in panel manufacturing, which has diminished, but in installation and maintenance. These "downstream" jobs highlight a key strength: the EU's ability to integrate imported components into value-adding services that drive local economies. The EU's targets for emissions reductions are more ambitious and nearer-term than those of China, positioning the continent to achieve cheap, clean energy relatively soon.

China's race to electrification

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The EU can lead in an electrostate transition by leaning into its strengths. As Garcia-Herrero and Mu show in a recent Bruegel analysis, China has become fully aware of the risks of stockpiling renewables in light of growing international protectionism and without the necessary investments in upgrading the power grid. Prioritising modernising Europe's power grid through smarter investments and cross-border infrastructure would mirror China's recent change in focus from green tech to green infrastructure and avoid this bottleneck.

The above optimism about Europe's decarbonisation strategy must be tempered with a critical eye towards Europe's growing dependence on China's green technology. China's dominance in this area – evident in its overwhelming share of global solar panel and battery production, and significant share of wind turbines while slapping tariffs on EVs to protect domestic industries. This inconsistency carries risks. Over-reliance on foreign imports exposes Europe to supply chain vulnerabilities, as seen during the 2022 energy crisis. At the same time, if China fully realises its electrostate vision, exporting green infrastructure components like grid upgrades or aluminium produced with cheap renewables could flood global markets with many more products than at present, further eroding European competitiveness.

The message is clear. Blind dependence on China isn't strategic autonomy; it's a recipe for deindustrialisation. Chinese firms like CATL and BYD are already building factories in Europe which create jobs but often transfer technology and profits back home. Ownership matters – Korean and Japanese investments have been less controversial, while China's state-backed model raises concerns over data security, intellectual property, and potential political leverage. Europe's tariffs on Chinese EVs are a start, but they address symptoms, not causes. Without bolstering domestic production, Europe could see its auto sector – which employs millions – hollowed out, much like solar manufacturing a decade ago. Emerging markets in Africa and South America, where price trumps brand, are already tilting toward Chinese EVs and sidelining European imports.

To navigate this, Europe must adopt a balanced strategy: embrace competition where it accelerates decarbonisation, like importing affordable equipment for a rapid solar rollout, but simultaneously investing aggressively in areas of competitive advantage and to reducing vulnerabilities. This means diversifying supply chains and enforcing stricter rules on foreign investment in critical sectors. In the same vein, the EU's Carbon Border Adjustment Mechanism could level the playing field so that high-emission imports are treated the same way as domestic producers.

Policymakers should heed warnings that China's early peak in emissions (potentially by 2027) isn't just environmental progress – it's a competitive edge. Europe must run faster, not slower, to secure cheap energy for its industries.

In conclusion, Europe's green tech story is one of untapped potential, not inevitable decline. By capitalising on its manufacturing capacities, job-creating services, and ambitious emissions targets, the continent can forge its own electrostate path, exporting high-value solutions worldwide. However, this requires vigilance against over-dependence on China, prioritising strategic autonomy to safeguard economic security. The choice is clear: compete boldly – and smartly –

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Authors

Alicia García-Herrero

Bruegel Senior Fellow

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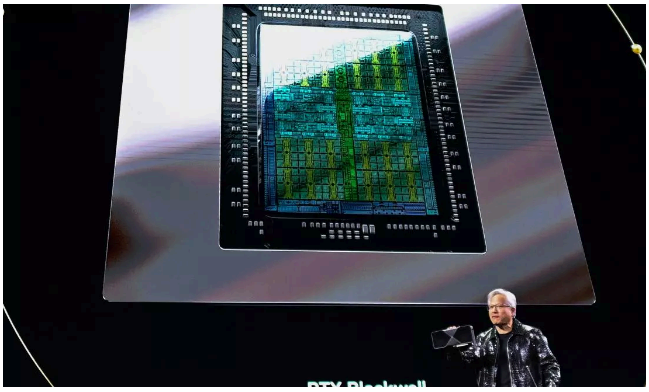

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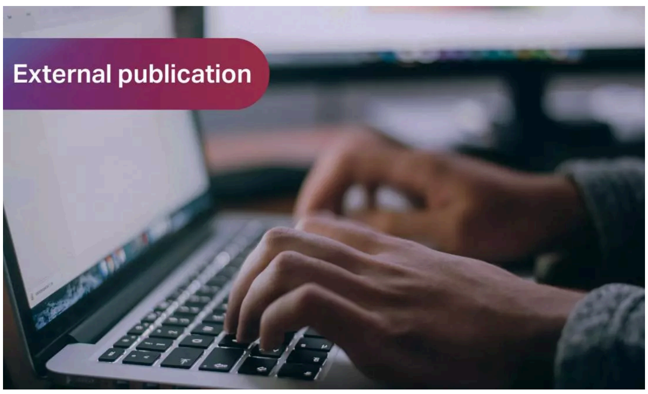

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